

Q1) Default Percentage

```
mysql> SELECT ROUND(AVG(Defaulted)*100,2) AS default_perc FROM loans;
+-----+
| default_perc |
+-----+
|          11.61 |
+-----+
```

Q2) Average Loan Amount & Income

```
mysql> SELECT ROUND(AVG(Income),0) AS avg_income,
-> ROUND(AVG(LoanAmount),0) AS avg_loan_amount
-> FROM loans;
+-----+-----+
| avg_income | avg_loan_amount |
+-----+-----+
|      82499 |       127579 |
+-----+-----+
```

Q3) Total Customers

```
mysql> SELECT COUNT(*) AS total_customers FROM loans;
+-----+
| total_customers |
+-----+
|          255347 |
+-----+
```

Q4) Income vs Default

```
mysql> SELECT
-> CASE
-> WHEN Income < 50000 THEN 'Low'
-> WHEN Income BETWEEN 50000 AND 75000 THEN 'Medium'
-> ELSE 'High'
-> END AS income_group,
-> ROUND(AVG(Defaulted)*100,2) AS default_perc
-> FROM loans
-> GROUP BY income_group
-> ORDER BY default_perc;
```

income_group	default_perc
High	9.35
Medium	10.65
Low	17.16

```
mysql> SELECT
-> CASE
-> WHEN Income < 50000 THEN 'Low'
-> WHEN Income BETWEEN 50000 AND 75000 THEN 'Medium'
-> ELSE 'High'
-> END AS income_group,
-> ROUND(AVG(Defaulted)*100,2) AS default_perc,
-> SUM(Defaulted) AS total_default,
-> COUNT(*) AS total_loans
-> FROM loans
-> GROUP BY income_group
-> ORDER BY default_perc;
```

income_group	default_perc	total_default	total_loans
High	9.35	13262	141882
Medium	10.65	5047	47369
Low	17.16	11344	66096

Q5) Credit Score vs Default

```
mysql> SELECT
-> CASE
-> WHEN CreditScore < 500 THEN 'Low'
-> WHEN CreditScore BETWEEN 500 AND 700 THEN 'Medium'
-> ELSE 'High'
-> END AS credit_group,
-> ROUND(AVG(Defaulted)*100,2) AS default_perc,
-> COUNT(*) AS total_loans
-> FROM loans
-> GROUP BY credit_group
-> ORDER BY default_perc;
```

credit_group	default_perc	total_loans
High	10.21	68946
Medium	11.49	93372
Low	12.77	93029

Q6) Loan Amount vs Default

```
mysql> SELECT
-> CASE
-> WHEN LoanAmount < 50000 THEN 'Low'
-> WHEN LoanAmount BETWEEN 50000 AND 150000 THEN 'Medium'
-> ELSE 'High'
-> END AS loan_group,
-> ROUND(AVG(Defaulted)*100,2) AS default_perc
-> FROM loans
-> GROUP BY loan_group
-> ORDER BY default_perc;
```

loan_group	default_perc
Low	8.06
Medium	10.30
High	14.52

Q7) Employment Type vs Default

```
mysql> SELECT
      -> EmploymentType,
      -> ROUND(AVG(Defaulted)*100,2) AS default_rate
      -> FROM loans
      -> GROUP BY EmploymentType
      -> ORDER BY default_rate DESC;
```

EmploymentType	default_rate
Unemployed	13.55
Part-time	11.97
Self-employed	11.46
Full-time	9.46

Q8) Education vs Default

```
mysql> SELECT
      -> Education,
      -> ROUND(AVG(Defaulted)*100,2) AS default_rate
      -> FROM loans
      -> GROUP BY Education
      -> ORDER BY default_rate DESC;
```

Education	default_rate
High School	12.88
Bachelor's	12.10
Master's	10.87
PhD	10.59

Q9) Income + Loan Combination Risk

```
mysql> SELECT
-> CASE
-> WHEN Income < 50000 THEN 'Low'
-> WHEN Income BETWEEN 50000 AND 75000 THEN 'Medium'
-> ELSE 'High'
-> END AS income_group,
->
-> CASE
-> WHEN LoanAmount < 50000 THEN 'Low'
-> WHEN LoanAmount BETWEEN 50000 AND 150000 THEN 'Medium'
-> ELSE 'High'
-> END AS loan_group,
->
-> ROUND(AVG(Defaulted)*100,2) AS default_rate
-> FROM loans
-> GROUP BY income_group,loan_group
-> ORDER BY default_rate DESC;
```

income_group	loan_group	default_rate
Low	High	24.13
Low	Medium	13.86
Medium	High	13.19
High	High	10.49
Medium	Medium	9.53
Low	Low	8.95
High	Medium	8.89
High	Low	7.84
Medium	Low	7.50

Q10) Credit Score + Income

```
mysql> SELECT
-> CASE
-> WHEN Income < 50000 THEN 'Low'
-> WHEN Income BETWEEN 50000 AND 75000 THEN 'Medium'
-> ELSE 'High'
-> END AS income_group,
->
-> CASE
-> WHEN CreditScore < 500 THEN 'Low'
-> WHEN CreditScore BETWEEN 500 AND 700 THEN 'Medium'
-> ELSE 'High'
-> END AS credit_group,
->
-> ROUND(AVG(Defaulted)*100,2) AS default_rate
-> FROM loans
-> GROUP BY income_group, credit_group
-> ORDER BY default_rate DESC;
```

income_group	credit_group	default_rate
Low	Low	18.74
Low	Medium	16.99
Low	High	15.27
Medium	Low	11.38
Medium	Medium	10.80
High	Low	10.46
Medium	High	9.48
High	Medium	9.17
High	High	8.08

Q11) DTI Ratio vs Default

```
mysql> SELECT
-> ROUND(DTIRatio,1) AS dti,
-> ROUND(AVG(Defaulted)*100,2) AS default_rate
-> FROM loans
-> GROUP BY dti
-> ORDER BY dti DESC;
```

dti	default_rate
0.9	12.40
0.8	12.26
0.7	12.32
0.6	11.85
0.5	12.01
0.4	11.26
0.3	11.23
0.2	10.62
0.1	10.48

Q12) Co-Signer Impact

```
mysql> SELECT
-> HasCoSigner,
-> ROUND(AVG(Defaulted)*100,2) AS default_rate
-> FROM loans
-> GROUP BY HasCoSigner;
```

HasCoSigner	default_rate
Yes	10.36
No	12.87

Q13) Mortgage Impact

```
mysql> SELECT
-> HasMortgage,
-> ROUND(AVG(Defaulted)*100,2) AS default_rate
-> FROM loans
-> GROUP BY HasMortgage;
```

HasMortgage	default_rate
Yes	10.88
No	12.35

Q14) Find Highest Risk Group (Most Important)

```
mysql> SELECT
-> EmploymentType,
-> Education,
-> ROUND(AVG(Defaulted)*100,2) AS default_rate
-> FROM loans
-> GROUP BY EmploymentType,Education
-> ORDER BY default_rate DESC
-> LIMIT 10;
```

EmploymentType	Education	default_rate
Unemployed	High School	15.05
Unemployed	Bachelor's	13.67
Part-time	High School	13.18
Self-employed	High School	13.01
Unemployed	Master's	12.94
Unemployed	PhD	12.54
Part-time	Bachelor's	12.53
Self-employed	Bachelor's	11.95
Part-time	Master's	11.41
Part-time	PhD	10.72